

Maintenance-free hybrid supercapacitors

Hybrid supercapacitors require no maintenance after installation. Data collected is securely sent to a cloud server typically located on-site, facilitating remote monitoring and management. This enables the receipt of instant alerts on critical issues such as temperature extremes or voltage irregularities without the need for physical checks. They eliminate the need for frequent maintenance visits. Unlike the maintenance-intensive lead-acid batteries, which often necessitate costly and challenging excursions to maintain, hybrid supercapacitors stand out by requiring no regular physical upkeep. This not only conserves resources but also ensures continual, effective management of energy storage, positioning hybrid supercapacitors as the preferred choice for powering essential equipment in hard-to-reach areas.

energy sources. These capabilities position hybrid supercapacitors not merely as alternatives to traditional batteries but as the next evolution in sustainable energy management.

A technological leap in energy management

Hybrid supercapacitors are transforming energy management by combining the rapid energy release capabilities of capacitors with the large storage capacity of batteries. This innovative technology enhances charging times and increases safety during energy discharge, establishing new benchmarks within the industry. The integration of a battery management system (BMS) is crucial, as it meticulously controls the storage and release of energy, ensuring optimal efficiency and safety. These devices leverage the best attributes of both capacitors and batteries, allowing for rapid charging through a process free from chemical reactions while storing substantial energy via a lithium battery anode.

One notable advancement in hybrid supercapacitors is their ability to mitigate the risk of dendrite growth—a prevalent issue in lithium batteries that can lead to short circuits and fires. Incorporating pure carbon, often graphene, in their cathodes, hybrid supercapacitors greatly reduce this risk, similar to advancements seen in synthetic motor oils, engineered for enhanced performance and durability.

Hybrid supercapacitors merge the physical attributes of capacitors with the chemical properties of batteries, creating a powerful solution tailored for contemporary power management challenges. They are designed to be versatile and dependable for a variety of applications, including telecommunications, residential use, and marine environments. In telecommunications, these supercapacitors are beginning to replace outdated lead-acid batteries, easily integrating into existing racking systems and providing substantial energy storage.

For residential applications, hybrid supercapacitors are available as wall units that offer between 6.5 to 14 kilowatt-hours of storage and are typically installed on the floor due to their weight. In marine settings, smaller, portable options like the Group 31 cell provide about 1.6 kilowatt-hours at 12 volts. Larger operations, such as those in the electric utility sector, utilise stacks of these modules managed by a master BMS, which can deliver up to about 850 volts when fully charged. Companies are actively scaling these technologies to accommodate larger capacities, with systems capable of handling up to three megawatts.

Gaining trust in hybrid supercapacitors

One of the main hurdles is customer acceptance. Companies, especially in fields like telecommunications and

utilities, tend to be cautious about adopting new technologies without a long track record. To address this, much effort has gone into distinguishing hybrid supercapacitors from the more familiar lithium-ion batteries through rigorous testing and certification, such as UL certification. These tests demonstrate the hybrid's superior performance, including better tolerance to high and low temperatures and longer cycle life, which significantly outperforms traditional lithium batteries.

Despite the initial scepticism, real-world applications are starting to show the advantages of hybrid supercapacitors. For example, in cell tower operations where they are cycled once or twice a day to complete discharge, hybrids have shown impressive performance and durability. These results are far better than what traditional lithium batteries could achieve under the same conditions.

Field testing by cable and telecom companies has further validated the resilience and efficiency of hybrid supercapacitors. When compared directly to lead-acid and lithium batteries, hybrids consistently outperform, surviving longer and performing better in cycle tests. This real-world proof is slowly shifting industry perception, building trust and understanding in the technology's unique benefits.

As more companies witness these tangible benefits, acceptance is growing, encouraging a broader adoption of hybrid supercapacitors. The technology is proving not just to be different, but superior, offering a compelling alternative to conventional energy storage systems.

Optimising performance in extreme conditions

Hybrid supercapacitors demonstrate exceptional versatility and reliability across a wide range of temperature